

7 SEM

1.

```
df_raw <- read.csv("data.csv")
df <- na.omit(df_raw)
head(df)
```

```
##   subj good_bad goodness q1 q2 q3 q4 q5 q6 q7 q8 q9 q10 q11 q12 q13 q14 q15
## 1    1         0      -3  2  2 -2 -1  1  1 -2 -2 -1  -2  -2  -1  -2  -2  -1
## 2    1         1       3 -2 -2  2  2 -2 -2  1  2  2  2  2  2  1  2  2
## 3    2         0      -2  1  2 -1 -2  2  1 -1 -2 -2  -1  -1  -2  -1  -1  -2
## 4    2         1       2 -1 -1  1  1 -1 -2  1  1  2  1  1  1  2  1  2
## 5    3         0      -3  2  2 -2 -2  2  2 -2 -2 -2  -2  -2  -2  -2  -2  -2
## 6    3         1       3 -2 -2  2  2 -2 -1  2  2  2  2  2  2  2  2  2
##   Gender Age
## 1      M  21
## 2      M  21
## 3      F  20
## 4      F  20
## 5      M  22
## 6      M  22
```

```
cat("      (N):", nrow(df))
```

```
##      (N): 42
```

2. 2

0.8 2 Positive 1 Barrier 2

- **Factor1 (Positive):** (q7-10) + (q11-13)
- **Factor2 (Barrier):** (q1, 2, 5, 6)

2

```
# q1 q15
df_fa <- df[, 4:18]

# =2
fa_result_2 <- factanal(df_fa, factors = 2, rotation = "promax")
print(fa_result_2, cutoff = 0.3)

##
## Call:
## factanal(x = df_fa, factors = 2, rotation = "promax")
##
## Uniquenesses:
##   q1    q2    q3    q4    q5    q6    q7    q8    q9    q10   q11   q12   q13
## 0.054 0.070 0.054 0.041 0.050 0.103 0.094 0.032 0.033 0.015 0.053 0.070 0.081
```

```
##    q14    q15
## 0.077 0.074
##
## Loadings:
##      Factor1 Factor2
## q1  -0.694  -0.308
## q2  -0.785
## q3   0.872
## q4           0.823
## q5  -0.821
## q6  -0.484  -0.498
## q7   0.517   0.470
## q8           0.839
## q9   0.977
## q10          0.942
## q11   0.595   0.412
## q12   0.355   0.641
## q13   0.748
## q14   0.642   0.350
## q15   0.516   0.482
##
##              Factor1 Factor2
## SS loadings      5.769   3.907
## Proportion Var   0.385   0.260
## Cumulative Var   0.385   0.645
##
## Factor Correlations:
##      Factor1 Factor2
## Factor1   1.000   0.862
## Factor2   0.862   1.000
##
## Test of the hypothesis that 2 factors are sufficient.
## The chi square statistic is 191.89 on 76 degrees of freedom.
## The p-value is 5.43e-12
```

3. SEM

2 goodness

```
#
model_improved <- '
#
#      "Positive"
Positive =~ q7 + q8 + q9 + q10 + q11 + q12 + q13
Barrier  =~ q1 + q2 + q5 + q6

#
goodness ~ Positive + Barrier
'

#
fit_improved <- sem(model_improved, data = df)
```

4.

P

```
# P
params <- parameterEstimates(fit_improved, standardized=TRUE)
reg_results <- subset(params, op == "~", select = c("lhs", "rhs", "std.all", "pvalue"))

colnames(reg_results) <- c(" ", " ", " ", " ", "P ")
print(reg_results)
```

```
##          P
## 12 goodness Positive      0.905 0.465
## 13 goodness  Barrier    -0.085 0.945
```

5.

```
#
fitMeasures(fit_improved, c("chisq", "df", "pvalue", "cfi", "rmsea", "srmr"))
```

```
##   chisq      df pvalue    cfi  rmsea  srmr
## 212.713 52.000  0.000  0.872  0.271  0.019
```

6.

```
semPaths(fit_improved,
  whatLabels = "std",
  layout = "tree",
  edge.label.cex = 1,
  curvePivot = TRUE,
  sizeMan = 5,
  sizeLat = 8)
```

